CHENJIE YANG

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GitHub Hugging Face

EDUCATION

Southwest Jiaotong University | Junior (3rd-year) | Ph.D. Applicant (Fall 2027)

- GPA: **3.95/4.0** | Rank: **1/220**
- Standardized Tests: TOEFL iBT: 100 | CET-6: 643 | CET-4: 595

RESEARCH INTERESTS

- World Models, Urban Scenario Modeling, Human Mobility Generation, and LLM-based Multi-Agent Systems

PUBLICATIONS

Under Review / Preprint

- (First author) *"ActivityEditor: Learning to Synthesize Physically Valid Human Mobility"* | ACL Rolling Review 2026 | Under Review | Corr.: Junbo Zhang, JD iCity | Project
- (Co-First author) *"RoboRouter: Training-Free Policy Routing for Robotic Manipulation"* | arXiv | Preprint | Corr.: Qingyao Wu, SCUT | Project
- (Co-First author) *"Automated Manipulation Learning"* | ICML 2026 | Under Review | Corr.: Qingyao Wu, SCUT | Project
- (Co-First author) *"UniTraj: Unified Trajectory-Guided Video Generation"* | IJCAI 2026 | Under Review | Corr.: Yuan Liu, CUHK

Accepted

- (Second author; advisor as first) *"Air-Ground Multimodal Transport Planning for Joint Passenger Mobility and Parcel Delivery: Integration of Drones, Aircraft, and Ground Vehicles"* | Transportation Research Part E | Accepted | Corr.: Haoning Xi, The University of Newcastle
- (First author) *"SARIMA-SVR Framework for Great Lakes Water Level Regulation"* | ICML 2025 Workshop NewInML | Accepted | Corr.: Chenjie Yang, SWJTU
- (Co-author) *"A Review of Optimization Methods for Low-Altitude Operations"* | Journal of Transportation Engineering and Information | Accepted | Corr.: Xiaobo Liu, SWJTU

Patent & Software Copyright

- Second inventor (advisor as first), *"A Multimodal Transport Planning Method for Low-Altitude and Ground Traffic Collaboration,"* pending.
- *"Cloud-based ShuMiBee Online Honey Sales Channel Management System,"* Software Copyright.

RESEARCH EXPERIENCE

RL-based Multi-Agent Negotiation for Activity Chain Generation

Tsinghua University | Intern | Project Leader Aug 2025–Mar 2026

Instructor: Dr. Junbo Zhang, Senior Researcher @ JD iCity

- **Planner–Editor scheduling with distillation**: built a dual-agent system; **distilled** an LLM teacher revision process into supervision traces and trained a **Qwen-Editor** with **SFT** and **GRPO**, achieving **30%+** accuracy improvement over baselines. Paper submitted to **ACL Rolling Review 2026**. ·(GitHub).
- **HoMe-LLM multi-agent activity chain generation**: proposed and implemented a **household-aware** and **negotiation-based** framework for **zero-shot** multi-person daily schedules; used **structured prompting + LoRA** on **NHTS 2017** cross-state diaries, and improved schedule generation accuracy by nearly **20%**; in preparation for **Transportation Research Part C**.

Intelligent Systems: Demand Forecasting and Multimodal Planning

Southwest Jiaotong University | **SRTP** | **Project Leader** May 2024–Apr 2025

Instructor: Dr. Haoning Xi, Senior Lecturer @ The University of Newcastle

- **Bike-sharing Rebalancing**: proposed a demand-driven rebalancing pipeline for urban mobility systems; developed **LSTM-based spatiotemporal demand forecasting** and coupled prediction with optimization to address **demand uncertainty** in routing and reduce operational cost.
- **Collaborative Air–Ground Path Planning**: formulated coupled routing for low-altitude vehicles and ground vehicles on hybrid air–ground networks; solved large-scale instances using **MILP + ALNS** with tailored constraints, analyzing efficiency–optimality trade-offs in **multimodal coordination**; paper accepted by **Transportation Research Part E**.

Generative Modeling and Style Alignment

Southwest Jiaotong University | **Project Leader** May 2025–Dec 2025

Instructor: Dr. Junkun Yuan, Research Scientist @ Tencent Hunyuan

- **BAGEL-based style transfer alignment**: built a **preference** alignment pipeline for text- and image-driven style transfer. Created a hierarchical style system and a **560-image** benchmark. Implemented automated evaluation with CLIP, LPIPS, and content loss, and integrated **Qwen3-VL** as a style recognizer achieving **95%+** accuracy. Trained with **DPO** under a flow-matching setup and stabilized efficient **2-A100** runs using **LoRA**.
- **UniTraj trajectory-guided video generation**: designed the UniTraj pipeline, led manuscript writing, and benchmarked key experiments. Paper submitted to **IJCAI 2026**.

RoboRouter and Automated Manipulation Learning

South China University of Technology | **Project Member** Aug 2025–Feb 2026

Instructor: Prof. Qingyao Wu, Associate Professor @ South China University of Technology

- **RoboRouter**: built a **training-free** router for natural-language manipulation tasks to dispatch among policies (**VLA/VA**); fine-tuned **pi0** and **ACT** on RoboTwin with **LoRA** and released checkpoints on HuggingFace and code (GitHub).
- **Automated Manipulation Learning**: developed a long-horizon agentic automation workflow with **RAG**, enabling task decomposition, tool use, and policy deployment. Paper submitted to **ICML 2026** (GitHub).

Enhancing Code Generation Robustness with LLMs

University of Cambridge | **Summer Program** | **Project Member** Jun 2024–Aug 2024

Instructor: Prof. Pietro Liò, Full Professor @ University of Cambridge

- Proposed an adaptive testing + self-reflection loop; used a debugging framework integrating reflection and **automated feedback** to solve robustness and accuracy issues in LLM-generated code.

AWARDS

- **National Scholarship**, 2024, 2025
- **National First Prize**, RAICOM Robot Developer Competition (RAICOM), 2025
- **National First Prize**, Electrician Mathematical Contest in Modeling (EMCM), 2024
- **Provincial First Prize**, Chinese Mathematics Competitions for College Students, 2024

SKILLS

- Technical Stack: **Programming**: Python, C++, C#, SQL | **Methods**: PyTorch, LoRA, SFT, DPO, GRPO, RAG | **Optimization**: MILP, ALNS | **DevOps**: Linux
- Certifications: Career Essentials in Generative AI (**Microsoft & LinkedIn**); NCRE Level 2 (C++); NCRE Level 2 (Python)

STUDENT LEADERSHIP

- Office Minister, Computer and Network Technology Association [2024–2025]
- Vice Minister, University Engineering Practice and Innovation Association